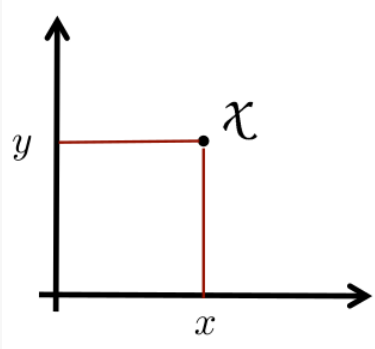
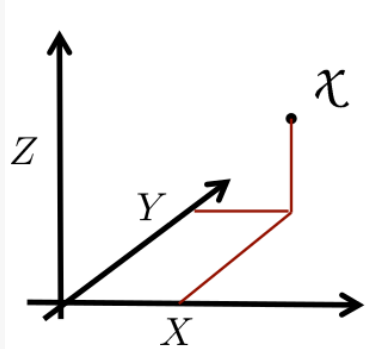
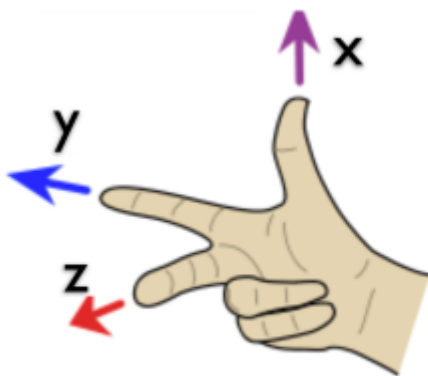


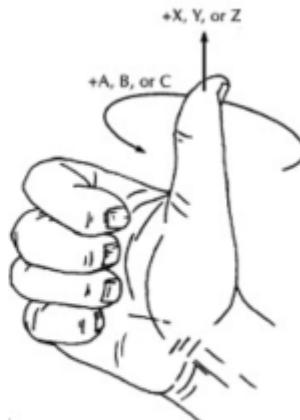
L1-1: Coordinate Transformations & Representations for Rotations

 $\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$	 $\mathbf{X} = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$
A Point in 2 D	A Point in 3 D

In 3D: Right-Handed CS



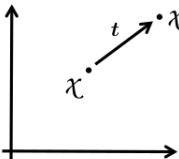
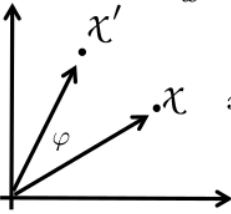
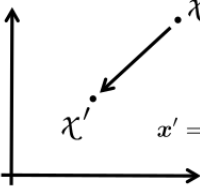
axes



positive rotation

Notation

- **Point** x (or y or p)
 - In Homogeneous coordinates $\mathbf{x}$
 - In Euclidian coordinates $\mathbf{x}$
- **2D** vs. **3D** space
 - Lowercase = 2D, Capitalised = 3D
- **Plane** $\mathcal{A}$
 - In homogeneous coordinates $\mathbf{A}$
- **Matrix** R
- **Quaternion** $\mathbf{q}$

Translation	Rotation	Scale Change
 $x' = \begin{bmatrix} x' \\ y' \end{bmatrix}$ $t = \begin{bmatrix} t_x \\ t_y \end{bmatrix}$ $x = \begin{bmatrix} x \\ y \end{bmatrix}$	 $x' = \begin{bmatrix} x' \\ y' \end{bmatrix}$ $x = \begin{bmatrix} x \\ y \end{bmatrix}$	 $x = \begin{bmatrix} x \\ y \end{bmatrix}$ $x' = \begin{bmatrix} x' \\ y' \end{bmatrix}$
$x' = \mathcal{T}(x) : \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$	$x' = \mathcal{R}(x) : \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$	$x' = \mathcal{D}(x) : \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \lambda x \\ \lambda y \end{bmatrix}$
Sum	Matrix Multiplication	Matrix Multiplication

Sum vs Matrix Multiplication

- No Single Operation to describe Transformation in Euclidian Space
 - Alternative representation as Quaternions
- Transformations in **Euclidian Space**
 - Combination of multiplication and additions
 - Rotation and Translation are **NOT** commutative
 - $\mathcal{T}(\mathcal{R}(x)) \neq \mathcal{R}(\mathcal{T}(x))$

Similarity Transformation

-	In 2D	In 3D
Scale Change	1 DoF	1 DoF
Rotation	1 DoF	3 DoF
Translation	2 DoF	3 DoF
Sum	4 DoF	7 DoF
-	$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{bmatrix} \begin{bmatrix} \lambda x \\ \lambda y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$	$\begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix} = \lambda R \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + \begin{bmatrix} t_X \\ t_Y \\ t_Z \end{bmatrix}$

Rotations

- Rotation is special transformation

$$\mathcal{R} : \mathbb{R}^n \rightarrow \mathbb{R}^N \quad \mathbf{x}' = R\mathbf{x}$$

- with $|R| = 1$ and $R^{-1} = R^T$
- **Properties**
 - Do not change the scale
 - Have fix point (rotation center)
- **Expression** through:
 - Rotation Matrices
 - Commonly used exchange format for rotations
 - Easy composition of rotations
 - Suboptimal for state estimation problems (9 parameters+constraints)
 - Euler angles
 - Minimal representation
 - useful for visualising rotations
 - Singularities & Discontinuities
 - No direct composition of rotations
 - Suboptimal for general state estimation problems
 - (Normalised) Axis-Angle Representation
 - Singularity for $\theta = 0$
 - Rotations limited to $(-\pi, \pi]$
 - Human readable
 - Normalised: Minimal representation
 - No direct composition of rotation (requires to build rotation matrix)
 - Quaternion
 - 4 Parameter presentation
 - Unique, except of sign
 - no singularities, no discontinuities
 - Easy composition of rotations
 - Allow for angular interpolation (SLERP)
 - Used for state estimation and handling rotations
 - *Matrix exponential of a skew-symmetric matrix*

3D Rotation Parameterisations

Representation	# Parameters	Unique	Readable	Singularities
Rotation Matrix	9	✓	✗	✗

Representation	# Parameters	Unique	Readable	Singularities
Euler Angles	3	✗	✓	✓
Rotation Axis and Angle	4(3)	(✗)	(✓)	(✓)
Quaternion	4(3)	✓	✗	✗

Rotation Matrix

- Rotation of the 2D plane by the angle φ

$$\circ R(\varphi) = \begin{bmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{bmatrix}$$

- Rotation in 3D: 3 axes to rotate

$$\circ R_x(\varphi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \varphi & -\sin \varphi \\ 0 & \sin \varphi & \cos \varphi \end{bmatrix}$$

$$\circ R_y(\varphi) = \begin{bmatrix} \cos \varphi & 0 & \sin \varphi \\ 0 & 1 & 0 \\ -\sin \varphi & 0 & \cos \varphi \end{bmatrix}$$

$$\circ R_z(\varphi) = \begin{bmatrix} \cos \varphi & -\sin \varphi & 0 \\ \sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- 9 DoF in general
- 6 independent constraints reduce degrees of freedom (each: -1 DoF)
 - > Overparameterised representation
- 3 degrees of freedom remain:

$$\circ \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} = [\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3] = [\mathbf{r}_1^T, \mathbf{r}_2^T, \mathbf{r}_3^T]$$

- Constraints:

- $|\mathbf{c}_1|^2 = 1$
- $|\mathbf{c}_2|^2 = 1$
- $|\mathbf{c}_3|^2 = 1$
- $\mathbf{c}_1^T \mathbf{c}_2 = 0$
- $\mathbf{c}_2^T \mathbf{c}_3 = 0$
- $\mathbf{c}_3^T \mathbf{c}_1 = 0$

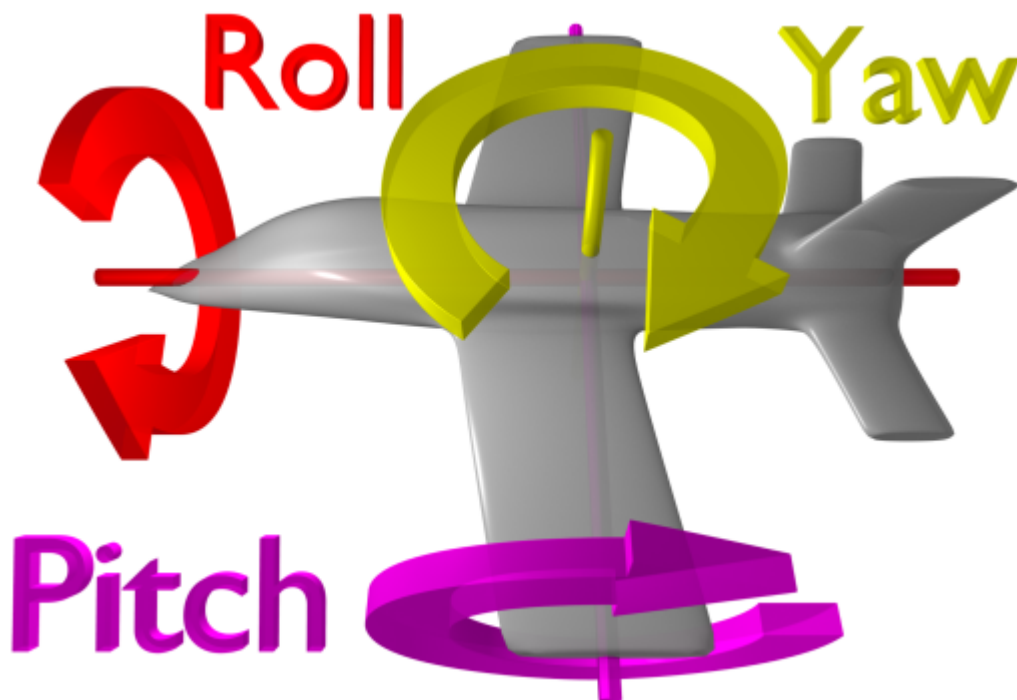
- Properties

- Column vectors $\mathbf{c}_i$ of R are the **images of the unit vectors** $\mathbf{e}_i$

- $\mathbf{c}_i = R\mathbf{e}_i = [\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3] \mathbf{e}_i$
- Column vector $\rightarrow$ previous unit vector
- Row vectors $\mathbf{r}_i$ of R are the vectors, which have been rotated into the unit vectors $\mathbf{e}_i$
- $\mathbf{r}_i^T = \mathbf{e}_i^T R = \mathbf{e}_i [\mathbf{r}_1^T, \mathbf{r}_2^T, \mathbf{r}_3^T]$
- Row vector $\rightarrow$ what is now a unit vector

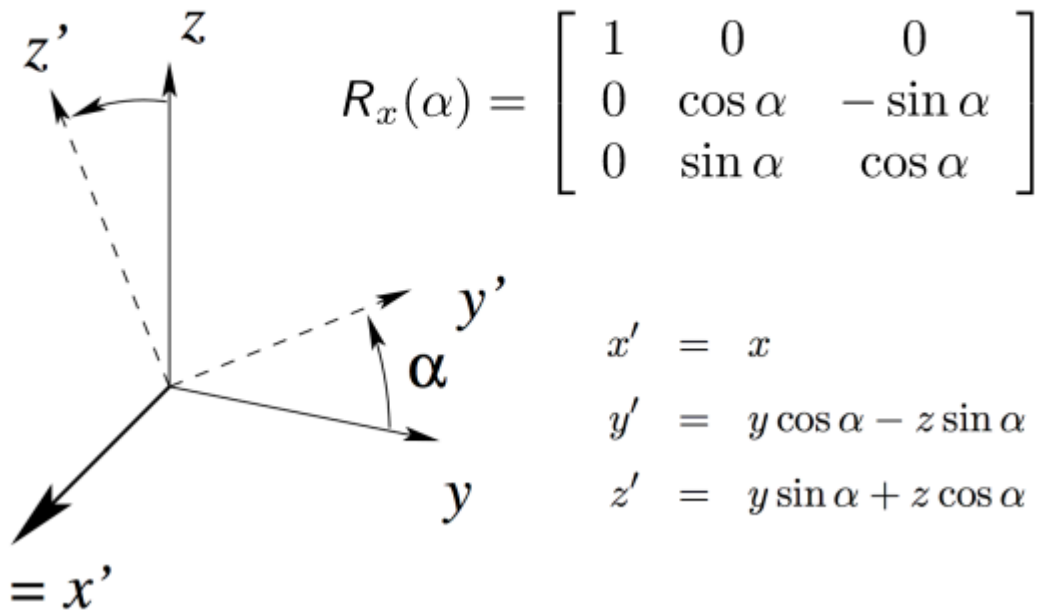
Euler Angles

- 3 rotations around fixed axes (e.g. z-y-x axes)
- Useful for visualising rotations
 - Commonly for robots, sensors, vehicle, planes
- 3 Variables for 3 degrees of freedom



- Different conventions though
- 3 Rotation Axes for 3 rotations
 - $R_1, R_2, R_3 \rightarrow X, Y, Z$
- e.g. for rotation around x-axis:

Rotation: $(\alpha, 0, 0)$



- The **Order** Makes a **Difference**

$$R_A(\alpha, \beta, \gamma) = R_3(\gamma)R_2(\beta)R_1(\alpha)$$

$$\begin{bmatrix} \cos \gamma \cos \beta & -\sin \gamma \cos \alpha + \cos \gamma \sin \beta \sin \alpha & \sin \gamma \sin \alpha + \cos \gamma \sin \beta \cos \alpha \\ \sin \gamma \cos \beta & \cos \gamma \cos \alpha + \sin \gamma \sin \beta \sin \alpha & -\cos \gamma \sin \alpha + \sin \gamma \sin \beta \cos \alpha \\ -\sin \beta & \cos \beta \sin \alpha & \cos \beta \cos \alpha \end{bmatrix}$$

$$R_B(\alpha, \beta, \gamma) = R_1(\alpha)R_2(\beta)R_3(\gamma)$$

$$\begin{bmatrix} \cos \gamma \cos \beta & -\sin \gamma \cos \beta & \sin \beta \\ \cos \gamma \sin \beta \sin \alpha + \sin \gamma \cos \alpha & -\sin \gamma \sin \beta \sin \alpha + \cos \gamma \cos \alpha & -\cos \beta \sin \alpha \\ -\cos \gamma \sin \beta \cos \alpha + \sin \gamma \sin \alpha & \sin \gamma \sin \beta \cos \alpha + \cos \gamma \sin \alpha & \cos \beta \cos \alpha \end{bmatrix}$$

Rotation Matrix to Euler Angles

- Given R : Compute (α, β, γ)
- Important: Specification for the convention of (α, β, γ)
 - Often 3-2-1: Yaw-Pitch-Roll OR
 - 1-2-3: Roll-Pitch-Yaw (ω, ϕ, κ)
- **Assumptions:**
 - Known axes and order of rotations
 - No **singularity** in Euler angle representation

$$R_A(\alpha, \beta, \gamma) = R_3(\gamma)R_2(\beta)R_1(\alpha) = \begin{bmatrix} \cos \gamma \cos \beta & -\sin \gamma \cos \alpha + \cos \gamma \sin \beta \sin \alpha & \sin \gamma \sin \alpha + \cos \gamma \sin \beta \cos \alpha \\ \sin \gamma \cos \beta & \cos \gamma \cos \alpha + \sin \gamma \sin \beta \sin \alpha & -\cos \gamma \sin \alpha + \sin \gamma \sin \beta \cos \alpha \\ -\sin \beta & \cos \beta \sin \alpha & \cos \beta \cos \alpha \end{bmatrix}$$

If $\cos \beta \neq 0$:

- $\alpha = \text{atan2}(R_{32}, R_{33})$
- $\beta = \text{atan2}(-R_{31}, \sqrt{R_{32}^2 + R_{33}^2})$
- $\gamma = \text{atan2}(R_{21}, R_{11})$

If $\cos \beta = 0$:

$$R_A = \begin{bmatrix} 0 & -\sin(\gamma - \alpha) & \cos(\gamma - \alpha) \\ 0 & \cos(\gamma - \alpha) & \sin(\gamma - \alpha) \\ -1 & 0 & 0 \end{bmatrix}$$

Singularity (Gimbal Lock)

- $\beta = \pm 90^\circ$, but α, γ not determinable
- Singularity at $\beta = 90^\circ$
- Illustration: Makes 3rd rotation axis equal to first one (except sign)
 - Thus: Only difference $\gamma - \alpha$ inferrable, but not α, γ

→ **Discontinuity!**

- Rotation angles limited to $[0, 2\pi)$
- "Wrap around"
- Values at borders can lead to parameter jumps
 - Problematic in state estimation

Axis-Angle Representation

- Result from Euler's Theorem
 - Every composition of 3D rotations expressable as
 - Single Rotation around Single rotation axis
- From Euler angles to Single rotation
- **4 parameters (3+1)** plus 1 constraint
 - Use Vector to encode **Rotation Axis** and
 - One Scalar for Rotation Angle
- $\theta, \mathbf{r} = [r_1, r_2, r_3]^T$ with $||\mathbf{r}|| = 1$
- Alternative Minimal Representation (3 parameters):

- $\mathbf{r} = [r_1, r_2, r_3]^T$ with $||\mathbf{r}|| = \theta$

Axis-Angle to Rotation Matrix

- Rodrigues Formula:

$$R_{\mathbf{r}, \theta} = \mathbf{I}_3 + \sin \theta \mathbf{S}_{\mathbf{r}} + (1 - \cos \theta) \mathbf{S}_{\mathbf{r}}^2$$

- With $\mathbf{S}_{\mathbf{r}} = \begin{bmatrix} 0 & -r_3 & r_2 \\ r_3 & 0 & -r_1 \\ -r_2 & r_1 & 0 \end{bmatrix}$

- Normalised:

- $R_{\theta} = \mathbf{I}_3 + \frac{\sin |\theta|}{|\theta|} \mathbf{S}_{\theta} + \frac{1 - \cos |\theta|}{|\theta|^2} \mathbf{S}_{\theta}^2$

- with $\theta = \theta \mathbf{r}$

Rotation Matrix to Axis-Angle

- $\theta = \text{atan2}(|\alpha|, \text{tr} R - 1)$

- with:

- $\alpha = - \begin{pmatrix} r_{23} - r_{32} \\ r_{31} - r_{13} \\ r_{12} - r_{21} \end{pmatrix} = 2 \sin \theta \mathbf{r}$

- $\text{tr}(R) = r_{11} + r_{22} + r_{33} = 1 + 2 \cos \theta$

- $\text{tr}(R)$: Trace - Sum of the main diagonal elements

- 3 cases for axis:

1. $\theta = 0 \leftrightarrow R = I$: Singularity

2. $\theta = \pi$: Sign of Axis is irrelevant:

- $\mathbf{r} \mathbf{r}^T = \frac{R+I}{2}$

- $\mathbf{r}$ given by any column of R

3. Otherwise $\mathbf{R} = \frac{\mathbf{a}}{|\mathbf{a}|}$

Quaternions

- Complex Number with 3-D complex component
- Only 4-parameter representation for rotation that:
 - **Unique**, except of the sign
 - **No singularities**
 - **No discontinuities**
 - *Almost minimal*

Definition

- Quaternion $\mathbf{q} = \begin{bmatrix} q \\ \mathbf{q} \end{bmatrix}$ is 4 D Vector with $\mathbf{q} = [q_1, q_2, q_3]^T$
- Can be interpreted as
 1. Complex number
 - 1D real number q and
 - 3D complex part $\mathbf{q} = q + q_1i + q_2j + q_3k$
 - with 3 complex units i, j, k
 - for which holds $i^2 = j^2 = k^2 = ijk = -1$
 2. Tuple
 - $\mathbf{q} = (q, \mathbf{q})$
 - Follow rules of certain algebra

Addition

- Element-wise sum (intuitively)

$$\begin{aligned} \mathbf{p} &= \mathbf{q} + \mathbf{r} \\ (p, \mathbf{p}) &= (q + r, \mathbf{q} + \mathbf{r}) \\ \begin{bmatrix} p_0 \\ p_1 \\ p_2 \\ p_3 \end{bmatrix} &= \begin{bmatrix} q_0 + r_0 \\ q_1 + r_1 \\ q_2 + r_2 \\ q_3 + r_3 \end{bmatrix} \end{aligned}$$

Multiplication

- Sum/difference between cross and dot-product
- Due to cross-product: **Not commutative**
- Unit quaternion is quaternion $\mathbf{q}$ with $|\mathbf{q}| = 1$

$$\begin{aligned} \mathbf{p} &= \mathbf{q}\mathbf{r} \\ (p, \mathbf{p}) &= (qr - \mathbf{q} \cdot \mathbf{r}, r\mathbf{q} + q\mathbf{r} + \mathbf{q} \times \mathbf{r}) \\ \begin{bmatrix} p_0 \\ p_1 \\ p_2 \\ p_3 \end{bmatrix} &= \begin{bmatrix} q_0r_0 - q_1r_1 - q_2r_2 - q_3r_3 \\ q_1r_0 + q_0r_1 + q_2r_3 - q_3r_2 \\ q_2r_0 + q_0r_2 - q_1r_3 + q_3r_1 \\ q_3r_0 + q_0r_3 + q_1r_2 - q_2r_1 \end{bmatrix} \end{aligned}$$

Inverse

- Inverse $\mathbf{q}^{-1}$ of quaternion
 - with conjugate $\mathbf{q}^*$:

$$\mathbf{q}^{-1} = \frac{\mathbf{q}^*}{|\mathbf{q}|^2}$$

$$= \frac{\begin{bmatrix} q \\ -\mathbf{q} \end{bmatrix}}{q_0^2 + q_1^2 + q_2^2 + q_3^2}$$

Rotations with Quaternions

$$\mathbf{q} = \begin{bmatrix} q \\ \mathbf{q} \end{bmatrix} = \begin{bmatrix} \cos(\theta/2) \\ \sin(\theta/2)\mathbf{r} \end{bmatrix}$$

- With normalised rotation axis:
 - $\mathbf{r} = [r_1, r_2, r_3]^T$ with $||\mathbf{r}|| = 1$
- Is Similar to Axis-Angle
 - $\theta, \mathbf{r} = [r_1, r_2, r_3]^T$ with $||\mathbf{r}|| = 1$

Example: Quaternion Rotation

- A rotation of θ around $\mathbf{r}$ with

$$\theta = 30^\circ \quad \mathbf{r} = \begin{bmatrix} 0.6 \\ 0.8 \\ 0.0 \end{bmatrix}$$

- Equation: $\mathbf{q} = \begin{bmatrix} q \\ \mathbf{q} \end{bmatrix} = \begin{bmatrix} \cos(\theta/2) \\ \sin(\theta/2)\mathbf{r} \end{bmatrix}$

- Result (in different notations):

$$\mathbf{q} = \begin{bmatrix} \cos(15^\circ) \\ \sin(15^\circ) \begin{bmatrix} 0.6 \\ 0.8 \\ 0.0 \end{bmatrix} \end{bmatrix}$$

$$\mathbf{q} = (\cos(15^\circ), \sin(15^\circ)(0.6, 0.8, 0.0))$$

$$\mathbf{q} = \cos(15^\circ) + \sin(15^\circ)(0.6i + 0.8j + 0.0k)$$

Executing a Rotation

- Perform rotation of vector part $\mathbf{p}$ of $p = (0, \mathbf{q})$
- Through left and right multiplication by $\mathbf{q} = (\cos(\theta/2), \sin(\theta/2)\mathbf{r})$

$$\mathbf{p}' = \mathbf{q}\mathbf{p}\mathbf{q}^{-1}$$

Composition

- (Multiple) Rotations as composition of quaternion multiplications

$$p' = q' p q'^{-1} \quad p'' = q'' p' q''^{-1}$$

$$\begin{aligned} p'' &= q'' (q' p q'^{-1}) q''^{-1} \\ &= (q'' q') p (q'^{-1} q''^{-1}) \\ &= \underbrace{(q'' q')}_{q} p \underbrace{(q'^{-1} q''^{-1})}_{q^{-1}} \\ &= q p q^{-1} \end{aligned}$$

- composite rotation $q = q'' q'$
 - Equal to $R_q = R_{q''} R_{q'}$
- of rotation quaternions q'', q'
- Execute rotation by

$$p'' q p q^{-1}$$

Quaternion and Rotation Matrix

- A unit quaternion can be transformed into a rotation matrix by

$$R_q = \begin{bmatrix} 1 - 2(q_2^2 + q_3^2) & 2(q_1 q_2 - q_0 q_3) & 2(q_1 q_3 + q_0 q_2) \\ 2(q_2 q_1 + q_0 q_3) & 1 - 2(q_1^2 + q_3^2) & 2(q_2 q_3 - q_0 q_1) \\ 2(q_3 q_1 - q_0 q_2) & 2(q_3 q_2 + q_0 q_1) & 1 - 2(q_1^2 + q_2^2) \end{bmatrix}$$

- Conversion in the other direction

$$\begin{aligned} q_0 &= \frac{1}{2} \sqrt{1 + \text{tr} R} \\ q_1 &= (R_{32} - R_{23}) / (4q_0) \\ q_2 &= (R_{13} - R_{31}) / (4q_0) \\ q_3 &= (R_{21} - R_{12}) / (4q_0) \end{aligned}$$

Summary

3D Rotation Parameterisations

Representation	# Parameters	Unique	Readable	Singularities
Rotation Matrix	9	✓	✗	✗
Euler Angles	3	✗	✓	✓
Rotation Axis and Angle	4(3)	(✗)	(✓)	(✓)
Quaternion	4(3)	✓	✗	✗

- Different Representations differ w.r.t.:
 - Readability
 - Singularities
 - Discontinuities
 - Minimal vs. overparameterised
 - Uniqueness
 - Simplicity of rotation composition
- -> No Best representation
- **Rotation Matrices**: Exchange Format
- **Euler Angles**: Visualisation
- **Quaternions**: State Estimation
- **Angle-Axis**: Readability and Link to Quaternions